

Mechanics - 1



Newton's Laws of Motion



Applications of Newton's Second Law



Forces in Newtonian Mechanics

Newton's First Law (Law of Inertia)

Statement

Newton's First Law: If no net force acts on a body ($\vec{F}_{\text{net}} = 0$), the body's velocity cannot change.

Meaning:

- An object at rest remains at rest.
- An object in motion continues with constant velocity in a straight line.

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Newton's Second Law (Law of Acceleration)

Statement

Newton's Second Law: The net force on a body is equal to the product of the body's mass and its acceleration.

$$\vec{F}_{\text{net}} = m\vec{a}$$

Where:

- $\vec{F}_{\text{net}}$: Net external force acting on the object
- m : Mass of the object (a measure of inertia)
- $\vec{a}$: Acceleration produced

Implication: Force and acceleration are *directly proportional*, while acceleration is *inversely proportional* to mass.

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Newton's Third Law (Action-Reaction Law)

Statement

Newton's Third Law: Every action has its equal and opposite reaction. If a body A exerts a force on body B , then body B exerts an equal and opposite force on body A .

$$\vec{F}_{A \rightarrow B} = -\vec{F}_{B \rightarrow A}$$

Important Points:

- Forces always occur in *pairs*.
- Action and reaction forces are equal in magnitude but opposite in direction.

Application of Newton's Second Law

Problem : 1

A box of mass $m = 0.20 \text{ kg}$ slides on a *frictionless ice surface* along the x -axis. The box experiences the following forces:

- $F_1 = 4.0 \text{ N}$ acting in the **positive x -direction** ($\rightarrow$),
- $F_2 = 2.0 \text{ N}$ acting in the **negative x -direction** ($\leftarrow$),
- $F_3 = 1.0 \text{ N}$ acting at an angle $\theta = 30^\circ$ above the x -axis.

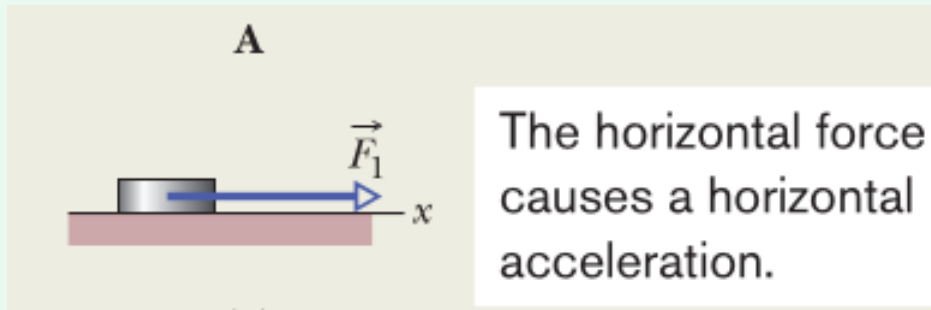
Findings:

Using Newton's second law

$$\vec{F}_{\text{net}} = m\vec{a}$$

Calculate the x -component of acceleration (a_x) in three different situations.

(a) Situation A: Single Force F_1 is acting



Only one force F_1 acts in the positive x -direction:

$$F_{\text{net},x} = F_1$$

Using Newton's second law:

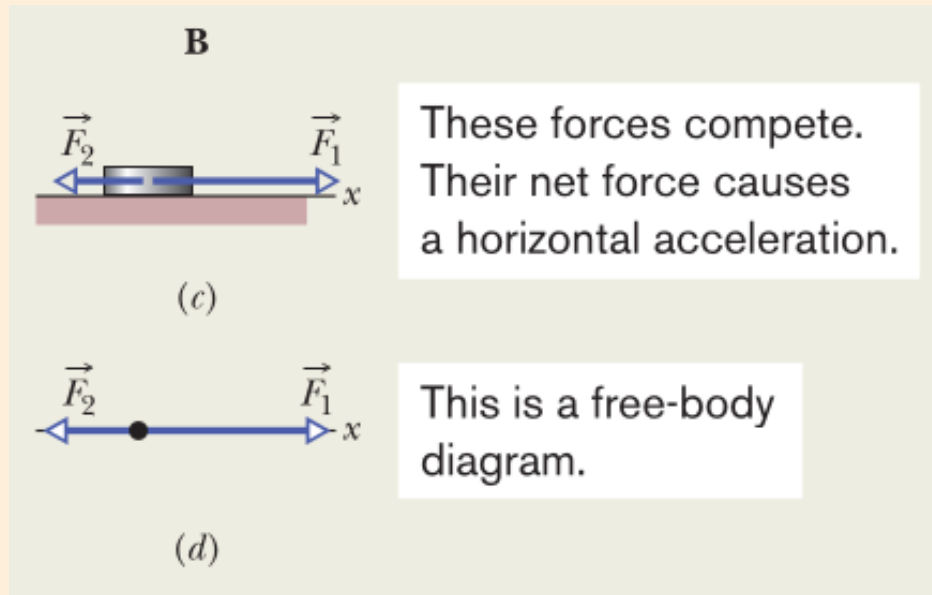
$$F_1 = ma_x$$

Solving for a_x :

$$a_x = \frac{F_1}{m} = \frac{4.0 \text{ N}}{0.20 \text{ kg}} = 20 \text{ m/s}^2$$

Thus, the acceleration is in the positive x -direction.

Situation B: Opposing Forces F_1 and F_2 are acting



Here, two forces act in opposite directions: F_1 in the positive x -direction and F_2 in the negative x -direction.

$$F_{\text{net},x} = F_1 - F_2$$

Applying Newton's second law:

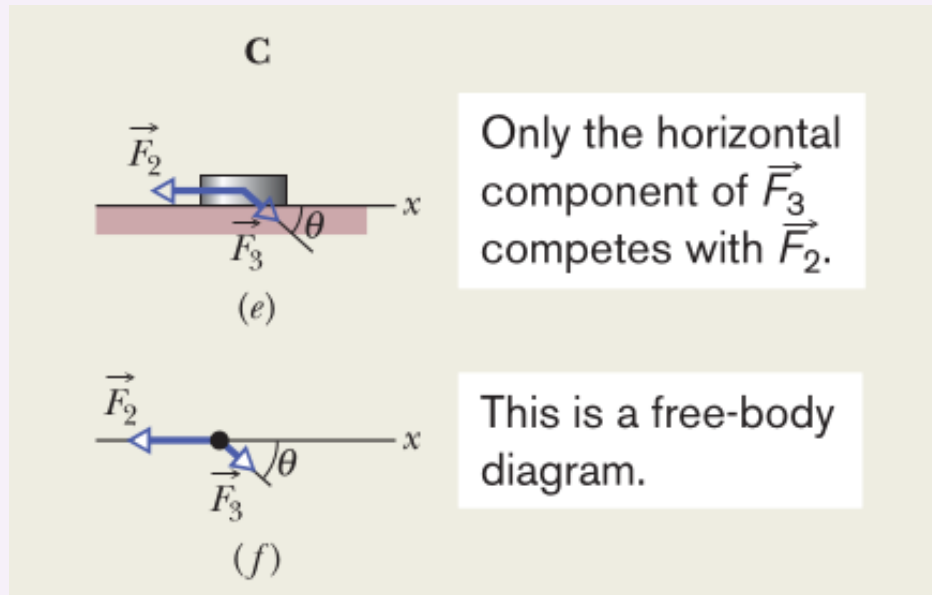
$$F_1 - F_2 = ma_x$$

Solving for a_x :

$$a_x = \frac{F_1 - F_2}{m} = \frac{4.0 \text{ N} - 2.0 \text{ N}}{0.20 \text{ kg}} = 10 \text{ m/s}^2$$

Thus, the acceleration is in the positive x -direction.

Situation C: Angled Force F_3 and F_2 are acting



Only the x -component of F_3 contributes to acceleration:

$$F_{3,x} = F_3 \cos \theta$$

Thus, the net force is:

$$F_{\text{net},x} = F_{3,x} - F_2$$

Applying Newton's second law:

$$F_3 \cos \theta - F_2 = ma_x$$

Solving for a_x :

$$a_x = \frac{F_3 \cos \theta - F_2}{m} = \frac{(1.0 \text{ N}) \cos 30^\circ - 2.0 \text{ N}}{0.20 \text{ kg}}$$
$$a_x = \frac{0.866 - 2.0}{0.20} = \frac{-1.134}{0.20} = -5.7 \text{ m/s}^2$$

The negative sign indicates that the acceleration is in the negative x -direction.

Forces in Newtonian Mechanics

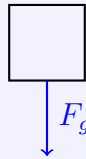
1. Gravitational Force F_g

Gravitational Force

The gravitational force pulls objects toward Earth's center.

$$F_g = mg$$

Object of mass m



Key Points:

- Direction: always **downward**.
- Magnitude proportional to m .
- Defines the **weight** of the object.
- $g \approx 9.8 \text{ m/s}^2$ near Earth.

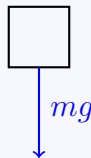
2. Free Fall

Free Fall

Only gravity acts on the object:

$$F_{\text{net},y} = ma_y$$

Free fall

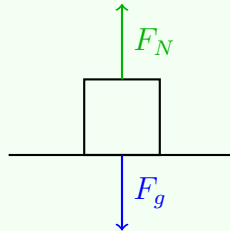


Implication: All objects fall with the same acceleration g , regardless of mass (ignoring air resistance).

3. Normal Force F_N

Normal Force

When a body presses against a surface, the surface pushes back with a **normal force**, always perpendicular to the surface.



$$F_N - F_g = ma_y$$

Key Points:

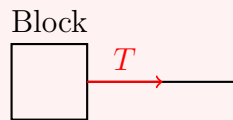
- Balances gravity when object is at rest.
- Perpendicular to the surface.
- Adjusts if object accelerates vertically.

4. Tension Force T

Tension

A taut cord pulls on objects with a **tension force**, directed **along the cord**, away from the object.

T = force transmitted by cord



Ideal Cord Assumptions:

- **Massless** : The cord is often treated as massless, meaning that its mass is negligible compared to the mass of the objects it connects.
- **Unstretchable**: The cord is often considered unstretchable, meaning that it does not stretch under the forces acting on it. This means the tension is assumed to be the same throughout the length of the cord, and there is no elongation or deformation of the cord itself.
- **Same tension throughout**.

5. Friction Force f

Friction

Friction is a resistive force opposing motion between surfaces.

$$f \leq \mu F_N$$

Types:

- **Static**: prevents motion ($f_s \leq \mu_s F_N$).
- **Kinetic**: opposes sliding ($f_k = \mu_k F_N$).

Direction: Always parallel to the surface and opposite to motion.